

A dispute agent that knows when not to act

AI-first intake for LATAM Bank card disputes in Spanish and Portuguese: it verifies the customer, finds the charge, files the dispute, and hands off to a person when policy says so.

100%

safe automated resolution (104/104) on a fresh sealed set, vs 86.5% for rules

74 / 74

required human handoffs made, with the right reason

0 / 230

unsafe outcomes; about \$0.0006 of LLM per resolved case

Disputes are where service fails, and where the data can be trusted

Contact reason	Share	1st-contact fix	Follow-up
Transactional	35.0%	92%	22%
Product	22.0%	90%	24%
Complaint	17.1%	44%	63%
Technical	15.0%	70%	41%

Complaint contacts take 431 s at the median, 2× a transactional one. Unrecognized charges and undue fees are ≈40% of the 67k formal complaints.

A ground truth to verify against

4.43M transactions, 0 orphans or owner mismatches: every named charge can be checked.

Isolation cannot be assumed

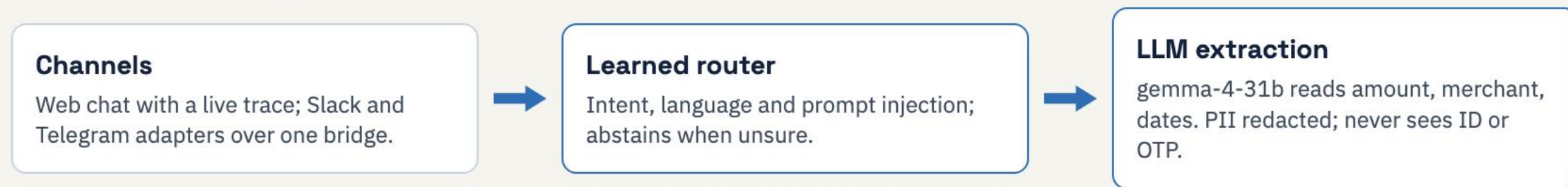
0 of 44,570 complaint products are owned by the complainant: enforce ownership in code.

No usable chat text

Transcripts are 2 blank templates, so the ES/PT intent set is team-labelled (declared).

The model understands; code decides

UNDERSTAND · learned, probabilistic



DECIDE AND ACT · deterministic, auditable



A write happens only if the session is verified, policy allows it, rules **and** model both read a clear “yes”, and the record reads back. Every turn is traced.

A learned router, chosen before the test set was opened

Test macro-F1 · 159 sealed, hand-written ES/PT utterances



Prompt-injection F1 on test: 1.00 for both learned routers, 0.91 for keywords. e5 won by the rule; **TF-IDF is deployed** because it fits the 512 MB free tier.

Labels

1,247 ES/PT utterances; the test set was written by hand and sealed in git first.

No leakage

Split grouped by seed; near-duplicates removed: 0 test↔train hits at cosine ≥ 0.9 .

Pre-registered

Selection rule fixed in advance; abstention threshold tuned on validation only.

Deployed choice checked

End to end, TF-IDF matched e5 on the held-out conversations (SAR 98.1–99.0% vs 99.0%).

230 sealed conversations on fresh records, same cases for both systems

Metric	Rules	Hybrid	Paired test
Safe automated resolution	86.5%	100%	$p = 0.0001$
Escalated when required	86.5%	100%	$p = 0.002$
Task success, Portuguese	80.9%	100%	$p < 0.001$
Unsafe outcomes	0 / 230	0 / 230	—
Unnecessary transfers	2.2%	0%	—
Turn latency p50 / p95	2.5 / 14 ms	1.7 / 9.0 s	—
LLM cost per resolution	\$0	\$0.0006	—

Deployed configuration, scored once on a new set sealed before any run. Before the last fix (v4): 98.1–99.0%, 1 unsafe.

How we got here

v2: 6 unsafe. The router alone picked the dispute reason.

v3: first fix regressed. Kept in the report.

v4: router and LLM disagree: ask. 1 unsafe left.

v5: rules and LLM read different reasons: ask. 0 unsafe.

Running today, and what it would take to make it real

Running now

- Live container on Render; bank data from a private dataset, startup fails closed
- Every turn traced: route, LLM tokens and cost, tool calls, policy rule
- Bounded retries; rules fallback when the LLM fails or hits its daily \$0.50 cap
- OTP with rate limit and lockout, signed sessions, idempotent writes
- 350 tests; make pipeline, router and eval rebuild every number

Honest gaps → next step

- Data is synthetic Spanish; PT is measured on our own labelled sets → real PT traffic
- Eval reuses message templates → new phrasings from real contacts
- The router often abstains in Portuguese → more labelled PT data, real traffic
- One process, in-memory sessions, SQLite → Postgres, Redis, real SMS and IdP
- Handoff is a JSON payload → an agent console and drift monitoring

Build something that works, prove it works, and know when not to act.